

Error for pennylane.numpy.tensor with plugin devices using non-differenti

<https://github.com/PennyLaneAI/pennylane/issues/900>

ROW 68

Error for pennylane.numpy.tensor with plugin devices using non-differentiable syntax #900

New issue

Closed

#903



antalszava opened on Nov 13, 2020

Issue description

When a `pennylane.numpy.tensor` is passed to a `QNode` as a keyword argument, it is considered as a non-differentiable parameter. Unlike in the case when it is handled as a differentiable parameter, it is not cast into a `ndarray`.

When indexing into a `pennylane.numpy.tensor`, one can end up having a single element sequence. Certain devices (e.g., `qiskit.aer`, `forest.wavefunction`), are not capable of supporting the application of a gate taking this type of argument and raise an error. Gates can take such a `pennylane.numpy.tensor` as gate parameter, when passed as a non-differentiable parameter.

```
import pennylane as qml
from pennylane.numpy import tensor

dev = qml.device("qiskit.aer", wires=4)

@qml.qnode(dev)
def circuit(phi=None):
    for y in phi:
        for idx, x in enumerate(y):
            qml.RX(x, wires=idx)
        return qml.expval(qml.PauliZ(0))

phi = tensor([[0.04439891, 0.14490549, 3.29725643, 2.51240058]])

circuit(phi=phi)
```

A special case using `RandomLayers` was tackled in PR #893. This is an example that is used in the [Quantumvolutional Neural Networks](#) demonstration.

- Expected behavior:**
The above snippet executes without errors.
- Actual behavior:**
Error is raised (Qiskit case):

```
~/anaconda3/lib/python3.7/site-packages/qiskit/providers/aer/backends/backend_utils.py in cpp_execute(circuit)
    49     qobj_dict['config']['library_dir'] = LIBRARY_DIR
    50
--> 51     return controller(qobj_dict)
    52
    53

RuntimeError: Invalid number of dimensions!
```

Further errors may arise for other devices (see [#893 \(comment\)](#) for error messages that were thrown when using `RandomLayers`).

- Reproduces how often:** (What percentage of the time does it reproduce?)
For certain plugin devices, each time a `pennylane.numpy.tensor` type sequence is passed as a gate parameter during execution.
- System information:**

```
Name: PennyLane
Version: 0.13.0.dev0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: https://github.com/XanaduAI/pennylane
Author: None
Author-email: None
```

Assignees

No one assigned

Labels

No labels

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

[Add tensor unwrapping in BaseQNode for keyword arguments](#)
PennyLaneAI/pennylane

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Participants



Name: PennyLane
Version: 0.13.0.dev0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: <https://github.com/XanaduAI/pennylane>
Author: None
Author-email: None
License: Apache License 2.0
Location:
Requires: numpy, scipy, networkx, autograd, toml, appdirs, semantic-version
Required-by: PennyLane-SF, PennyLane-Qchem, PennyLane-Forest, PennyLane-qsharp, pennylane-qulacs, PennyLane-li
Platform info: Linux-4.19.11-041911-generic-x86_64-with-debian-buster-sid
Python version: 3.7.6
Numpy version: 1.18.5
Scipy version: 1.4.1
Installed devices:
- strawberryfields.fock (PennyLane-SF-0.12.0)
- strawberryfields.gaussian (PennyLane-SF-0.12.0)
- strawberryfields.gbs (PennyLane-SF-0.12.0)
- strawberryfields.remote (PennyLane-SF-0.12.0)
- strawberryfields.tf (PennyLane-SF-0.12.0)
- forest.numpy_wavefunction (PennyLane-Forest-0.13.0.dev0)
- forest.qpu (PennyLane-Forest-0.13.0.dev0)
- forest.qvm (PennyLane-Forest-0.13.0.dev0)
- forest.wavefunction (PennyLane-Forest-0.13.0.dev0)
- microsoft.QuantumSimulator (PennyLane-qsharp-0.9.0.dev0)
- qulacs.simulator (pennylane-qulacs-0.12.0.dev0)
- lightning.qubit (PennyLane-lightning-0.10.0.dev0)
- qiskit.aer (PennyLane-qiskit-0.13.0.dev0)
- qiskit.basicaer (PennyLane-qiskit-0.13.0.dev0)
- qiskit.ibmq (PennyLane-qiskit-0.13.0.dev0)
- default.gaussian (PennyLane-0.13.0.dev0)
- default.mixed (PennyLane-0.13.0.dev0)
- default.qubit (PennyLane-0.13.0.dev0)
- default.qubit.autograd (PennyLane-0.13.0.dev0)
- default.qubit.tf (PennyLane-0.13.0.dev0)
- default.tensor (PennyLane-0.13.0.dev0)
- default.tensor.tf (PennyLane-0.13.0.dev0)
- aqt.noisy_sim (PennyLane-AQT-0.10.0.dev0)
- aqt.sim (PennyLane-AQT-0.10.0.dev0)
- cirq.mixedsimulator (PennyLane-Cirq-0.12.0.dev0)
- cirq.simulator (PennyLane-Cirq-0.12.0.dev0)

Source code and tracebacks

Additional information

A solution could be to create a case for `pennylane.numpy.tensor` type sequence parameters in each plugin device implementation (see [#893 \(comment\)](#) containing a suggestion).

Seems that this problem does not arise in tape mode.



josh146 on Nov 13, 2020

Member ...

Thanks for catching this @antalszava @mariaschuld.

I think this is a result of the old + new QNode variable marking system clashing.

- Keyword arguments being non-differentiable is only assumed in the old core; PennyLane NumPy tensors are passed as-is to the device if passed as a keyword argument.
- In tape mode, however, positional vs. keyword arg makes no difference. Any argument can be differentiable, if marked as so.

As a result, when tape-mode is default this issue should disappear. However, in the meantime, I suggest we add

```
if isinstance(x, qml.numpy.tensor) and x.ndim == 0:  
    x = x.unwrap()
```

either to the (old) autograd interface, the (old) `BaseQNode`, or `QubitDevice`, depending on which one makes more sense; this way we don't have to update all plugins.

```
- aqt.noisy_sim (PennyLane-AQT-0.10.0.dev0)
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My gut feeling is that it makes more sense to fix this in `BaseQNode`, since it will be deleted anyway when tape mode is default.



[@antalszava](#) mentioned this on Nov 14, 2020

🔗 [Add tensor unwrapping in BaseQNode for keyword arguments #903](#)



[@antalszava](#) closed this as `completed` in [#903](#) on Nov 16, 2020

Add tensor unwrapping in BaseQNode for keyword arguments #903

<> Code

Merged antalszava merged 10 commits into `master` from `non_diff_tensor_fix` on Nov 16, 2020

Conversation 11 Commits 10 Checks 0 Files changed 5

+88 -7



antalszava commented on Nov 14, 2020 • edited

Contributor

Context:

- 🔗 [Error for pennylane.numpy.tensor with plugin devices using non-differentiable syntax #900](#)
- 🔗 [Unwrap tensor in random_layer #893](#)

Description of the Change:

Adds tensor unwrapping in BaseQNode for keyword arguments.

Benefits:

A more general solution compared to [#893](#) for handling `pennylane.numpy.tensor` objects as gate parameters.

Related PennyLane-Qiskit tests:

- 🔗 [Test PennyLane tensor unwrapping pennylane-qiskit#115](#)

Possible Drawbacks:

N/A

Related GitHub Issues:

Closes [#900](#)



antalszava added 3 commits 5 years ago

- 🔗 [Fix in BaseQNode](#)
- 🔗 [Add staticmethod for unwrapping \(codefactor\); remove import](#)
- 🔗 [Change to check that spots init_state difference earlier](#)

Reviewers

josh146



Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

🔗 [Error for pennylane.numpy.tensor with plugi...](#)

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